

# Place Value, Rounding and Estimating — Paper A

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to be handled on paper.
- The decimal point does not move. The **digits** move: one column left for each  $\times 10$ , one column right for each  $\div 10$ .
- $\times 0.1$  is the same as  $\div 10$ , and  $\div 0.1$  is the same as  $\times 10$ . Rewrite the question that way before you calculate.
- Read every rounding instruction twice: **decimal places** and **significant figures** give different answers on the same number.
- Write a zero before the decimal point (0.47, never .47), and keep placeholder zeros (38 000, never 38).
- The mark for each question is shown in square brackets on the right.

SECTION 1 – POWERS OF TEN, AND 0.1

1. Complete the table. Each empty box is worth one mark. [6]

| Number | $\times 100$ | $\div 100$ |
|--------|--------------|------------|
| 5.3    |              |            |
| 47     |              |            |
| 0.6    |              |            |

Count the zeros to count the columns. The point stays put; the digits travel.

2. Work out:

(a)  $8.2 \times 10$  [1]

(b)  $3.5 \div 100$  [1]

(c)  $0.09 \times 1000$  [1]

**(d)**  $620 \div 10\,000$

[1]

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**3.** Work out:

**(a)**  $7.4 \times 0.1$

[1]

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**(b)**  $5 \div 0.1$

[1]

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**(c)**  $3.8 \times 0.01$

[1]

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**(d)**  $6 \div 0.01$

[1]

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SECTION 2 – ROUNDING

**4.** Round **4736** to:

**(a)** the nearest 10

[1]

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**(b)** the nearest 100

[1]

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**(c)** the nearest 1000

[1]

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**(d)** 1 significant figure

[1]

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**5.** Round **12.3648** to:

(a) 1 decimal place [1]

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(b) 2 decimal places [1]

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(c) 3 decimal places [1]

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(d) 2 significant figures [1]

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6. Round each of these to **2 significant figures**.

(a) 0.06382 [1]

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(b) 5471 [1]

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(c) 0.9047 [1]

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(d) 38.72 [1]

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SECTION 3 – ESTIMATING

7. Estimate  $61.4 \times 4.8$  by rounding each number to 1 significant figure. Show what you rounded each number to. [2]

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- 8.** Estimate  $3891 \div 19$  by rounding each number to 1 significant figure. Show what you rounded each number to. [2]

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*End of paper. Check every answer has a unit or a form the question actually asked for.*